Katherine Ormerod

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RESEARCH INTERESTS

High-redshift galaxies, galaxy formation and evolution, cosmic reionisation, dust formation and properties, galaxy morphology, multi-wavelength astronomy, resolved studies of galaxies, dust and metal enrichment

EDUCATION

Astrophysics PhD, 2023 - Present
Astrophysics Research Institute, Liverpool John Moores University
Thesis: *A Multi-Wavelength View of Dust and Star Formation in Reionisation-Era Galaxies*
Supervisors: Dr. Renske Smit (LJMU), Dr. Joris Witstok (DAWN)

Astronomy & Astrophysics MSc by Research, 2022 - 2023
Jodrell Bank Centre for Astrophysics, University of Manchester
Thesis: *The Size and Shape Evolution of Galaxies using the James Webb Space Telescope*
Supervisor: Prof. Christopher J. Conselice

Physics, Astrophysics and Cosmology BSc (Hons), 2018 - 2021
Lancaster University
Thesis: *Investigating the properties of potential metal poor stars in the Milky Way*
Supervisor: Dr. David Sobral

SELECTED PUBLICATIONS

22 total publications, including **4** first author, **967** total citations (173 to first-author publications), **h-index=15**. For a full list of publications, see my SciX link.

First Author:

1. IRAC-Excess Galaxies in the NIRSpec Wide Survey: physical properties of bright EELGs at $z \sim 7$ and implications for Reionization, **Katherine Ormerod**, Renske Smit, Joris Witstok et al., 2026, submitted to MNRAS, arXiv:2610.20332
2. NOEMA probes the [CII] and dust content in a 2175Å UV Bump Galaxy at $z = 7.1$, **Katherine Ormerod**, Renske Smit, Joris Witstok et al., 2026, submitted to MNRAS, arXiv:2609.12203
3. Detection of the 2175Å UV bump at $z > 7$: evidence for rapid dust evolution in a merging reionization-era galaxy, **Katherine Ormerod**, Joris Witstok, Renske Smit, et al. 2025, MNRAS, 542, 1136–1154
4. EPOCHS VI: the size and shape evolution of galaxies since $z \sim 8$ with JWST observations, **Katherine Ormerod**, Christopher J. Conselice et al., 2024, MNRAS, 527, 6110-6125 (**>130 citations**, Web of Science Highly Cited Paper 2023)

Selected Co-authored Papers:

1. Detection of [OIII] 88μm in JADES-GS-z14-0 at $z = 14.1793$, Sander Schouws, Rychard J. Bouwens, **Katherine Ormerod**, et al. 2025, ApJ, 988, 19.
   Led analysis of CLOUDY models that yielded a robust lower limit on the oxygen abundance.
2. Galaxy Quenching at the High Redshift Frontier: A Fundamental Test of Cosmological Models in the Early Universe with JWST-CEERS, Asa Bluck, Christopher J. Conselice, **Katherine Ormerod**, et al. 2024, ApJ, 961, 163.
   Led morphological analysis of galaxy sample through surface brightness fitting.

SUCCESSFUL OBSERVING PROPOSALS

5 principal investigator (∼84 hours), 8 co-investigator

Principal investigator:

- 'Constraining Dust Formation at Cosmic Dawn with Luminous UV Bump Galaxies', **JWST**/NIRSpec IFU, 19.1 hours
- 'Constraining the dust properties of a luminous UV bump galaxy within the Epoch of Reionisation', 2026.1.01430.S, **ALMA**, 17.5 hours
- 'A Precious Glance into the Low-Mass Building Blocks of Galaxies at the Earliest Times', 2026.1.01449.S, **ALMA**, 6 hours
- 'The 2175Å UV bump as a signpost of NIRCam dark star formation', **NOEMA** interferometer, 25 hours
- 'Unveiling dust build up in a 'UV bump' galaxy at z=7.1 with JWST+NOEMA', **NOEMA** interferometer, 16 hours (Co-PI)

Co-Investigator:

- 'Efficiently unveiling luminous dust reservoirs in the first billion years', 2026.1.00854.S, ALMA, 38.9 hours
- 'To peak or not to peak? Molecular Gas at Cosmic Dawn: ALMA Band 2 Observations of REBELS-25', 2026.1.00848.S, ALMA, 9.7 hours
- 'Confirming surprisingly high metallicities in galaxies in the Epoch of Reionization', JWST/NIRSpec IFU, 18.9 hours
- 'Pushing the Dust Frontier: Characterisation of the Most Distant Dusty Galaxy', JWST/NIRSpec IFU, 1.8 hours
- 'Using [NIII]57 to constrain N/O in GSz14: uncovering the origin of ultra-luminous galaxies at $z > 10$', 2025.1.01269.S, ALMA, 25.4 hours
- 'Confirming Three Hidden ULIRGs in the Epoch of Reionization', 2025.1.00232.S, ALMA, 9.6 hours
- 'Unveiling the ISM at z>14: A Critical Study of [CII]158μm in GS-z14, 2024.A.00007.S, ALMA, 19.3 hours
- 'A measurement of [N/O] abundance at z=12.3: on the origin of ultra-luminous galaxies at cosmic dawn', 2024.1.01645.S, ALMA, 7.9 hours

CONFERENCE TALKS

Invited Talks

2026 *MPIA Galaxy Coffee Seminar*, MPIA, Heidelberg, Germany - **Invited seminar**
2026 *British Federation of Women Graduates North West Academic Presentation Day* - **Invited speaker**
2026 *Exploring the Aromatic Universe in the JWST Era*, London, Ontario, Canada - **Invited highlight talk:** "JWST observations of the 2175Å UV Bump at high redshift"
2026 *Miracles of the Early Universe Meeting II*, Geneva, Switzerland - **Invited speaker**
2026 *From Dust till Dawn*, Leiden, Netherlands - **Invited speaker**
2025 *REBELS Collaboration Meeting*, Valencia, Spain - **Invited speaker**

Contributed Talks

2026 *Manchester–Liverpool–Lancaster Extragalactic Meeting*, Manchester, UK
2026 *Charting Cosmic Dawn in Copenhagen*, Copenhagen, Denmark (flash talk)
2025 *First Galaxies*, Oxford, UK
2024 *Dust and Gas Throughout Cosmic Time*, Hiroshima, Japan (virtual)
2024 *National Astronomy Meeting*, Hull, UK
2024 *Extreme Galaxies in Their Extreme Environments at Extremely Early Epochs*, Reykjavik, Iceland
2024 *Raising the Veil on Star Formation Near and Far*, Cambridge, UK
2023 *National Astronomy Meeting*, Cardiff, UK

HONOURS AND FUNDING

2026 British Federation of Women Graduates North West Travel Bursary (£500)

2025-26 LJMU Postgraduate Conference Travel Fund (£400)
2023 Web of Science Top 1% Cited Publication in Astronomy & Astrophysics during 2023 for *EPOCHS VI: the size and shape evolution of galaxies since $z \sim 8$ with JWST observations*
2018 Lancaster Scholarship for academic achievement (£2000)

ACADEMIC SERVICE AND OUTREACH

- Member of the Scientific Organising Committee for the NAM 2025 Session **"Forging the elements: Understanding chemical evolution and stellar populations across cosmic time"**.
- Reviewer for the **Astrophysical Journal**.
- Reviewer for **Astronomy & Astrophysics**.
- Deliver outreach talks aimed at 11-18 year old students in local schools (2023-Present)
- Delivered a public talk at Research Roasters as part of the 2024 British Science Festival
- Lancaster University Physics Department Ambassador, 2018-2019

TEACHING AND SUPERVISION

Supervision

2025-26 Co-supervision of Master's student searching for the 2175Å UV bump feature in archival data
2025-26 Co-supervision of three BSc students
2024-25 Co-supervision of two BSc students

Teaching

2026-27 Graduate teaching assistant for undergraduate Galaxies module - LJMU
2025-26 Graduate teaching assistant for undergraduate Galaxies module - LJMU
2024-25 Graduate teaching assistant for undergraduate Introduction to Astrophysics module - LJMU
2022-23 Graduate lab demonstrator for undergraduate astronomy labs - University of Manchester

DATA REDUCTION

- Extensive experience reducing and imaging NOEMA data using GILDAS and MAPPING
- Extensive experience reducing ALMA data, producing spectral line cubes with CASA
- Experience with JWST NIRSpec/IFU pipeline, attended ESA JWST data reduction masterclass

OTHER TRAINING

2026 2026 ALMA Workshop, Manchester, UK - *Focused on ALMA data reduction, Next Generation Observing Tool, ALMA Wideband Sensitivity Upgrade.*
2025 2nd IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution, Granada, Spain - *Advanced lectures on observational and theoretical galaxy formation, including poster presentation.*
2023 ESA JWST Data Masterclass, Madrid, Spain - *Training covering JWST reduction pipeline, including hands-on workshops.*

MEDIA COVERAGE

2025 "Oxygen discovered in most distant early galaxy" - Helped coordinate LJMU press release to accompany official ESO and ALMA press releases. Also covered by news outlets including the BBC Sky at Night and CNN.